Mesh Quality Report

Oct 29, 2026 · @VENKATA RAMA ADITYA PHILKHANA

Summary

All 10 meshes are valid. All 20 regions pass OpenFOAM checkMesh ("Mesh OK"), and none has an inverted cell. Since 5 October the solid is meshed on the fluid's atom grid, so fluid and solid share their interface faces. The interface heat error is now 0.000 % on the four pin-fins and 0.036 % on the gyroid, and 99.99–100.03 % of the heater power reaches the water. Every layered mesh reaches a mean y+ below 1. Of the atom-test meshes, the square at 0.45 and 0.2 mm passes checkMesh; the gyroid at 0.15 mm fails it in the solid on skewness (section Atom-test meshes). Two points remain open: non-orthogonality within 1° of the 70° limit in two TPMS fluids, and mesh independence (atom test planned). The wall-layer test (6 layers, jump to the core cell 2.3 instead of 4.7) changed the peak heater temperature by less than its oscillation band, so the growth ratio and the layer-to-core jump are accepted with a stated uncertainty.

The campaign covers 4 pin-fin plates (0.3 mm atoms, fluid with 5 wall layers) and 3 TPMS plates (0.2 mm atoms), the TPMS each with and without wall layers. Fluid and solid are two regions of one OpenFOAM v2312 conjugate heat transfer case. The pin-fin solids (`_M`) and the layered TPMS solids are meshed on the fluid's grid, with their outer walls moved onto the CAD. The TPMS meshes without layers keep separately meshed solids; they are reference runs and were not re-meshed. The 9 October version replaced the earlier solids with the final ones and added the final energy balances; this version (13 October) adds the wall-layer test.

Criterion	Pass value	Pin-fins (4)	TPMS, layers, matched (3)	TPMS, no layers (3)
Inverted cells	0	Pass (0)	Pass (0)	Pass (0)
Min scaled Jacobian	> 0	Pass (fluid 0.060–0.075, solid 0.203–0.443)	Pass (fluid 0.050, solid 0.200–0.205)	Pass (0.263–0.268)
checkMesh verdict	Mesh OK	Pass, all 8 regions	Pass, all 6 regions	Pass, all 6 regions
Max skewness (OpenFOAM)	⩽ 4	Pass (3.44–3.56)	Pass (3.56)	Pass (⩽ 0.80)

Criterion	Pass value	Pin-fins (4)	TPMS, layers, matched (3)	TPMS, no layers (3)
Max non-orthogonality	⩽ 70°	Pass (49.9–57.2°)	Pass, narrow (67.0–69.5°)	Pass (⩽ 26.2°)
Max aspect ratio	⩽ 1000	Pass (⩽ 47.9)	Pass (⩽ 102.9)	Pass (⩽ 2.40)
Mean y+	< 1	Pass (0.40–0.60)	Pass (0.57–0.59)	Fail (9.9–10.3), by design
Growth ratio	⩽ 1.2	Above (1.30)	Above (1.30)	No layers
Layer coverage	⩾ 99 %	Pass (100 %)	Pass (99.8–100 %)	No layers
Mean wall deviation	⩽ 0.25 × atom	Pass (fluid 4–7 %, solid outer walls < 1 %)	Pass (fluid 17–20 %, solid outer walls < 1 %)	Pass (6–15 %)
Volume deviation	⩽ 1 % fluid, ⩽ 2 % solid	Pass (⩽ 0.20 / ⩽ 0.32 %)	Pass (⩽ 0.12 %)	Pass (⩽ 0.06 %)
Interface area match	⩽ 1 %	Pass (0.00 %)	Pass (⩽ 0.06 %)	Not measured
Energy balance	Within 1 %	Pass (99.991–99.996 %)	Pass (100.03–100.12 %)	2 of 3 (Lidinoid 98.65 %)
Interface heat error	Within 1 %	Pass (0.000 %)	Pass (0.04–0.13 %)	2 of 3 (Lidinoid −1.34 %)

Percentages for wall deviation are the mean deviation as a share of the atom size. For the layered TPMS, the Lidinoid and Neovius solution values come from the previous code version (5 October); their final-code meshes pass the same mesh checks, but their re-runs were not completed. The sections below give the values for each design.

Criteria and pass thresholds

Each mesh is judged on 17 criteria. Nine come from OpenFOAM, Ansys or published CFD guidance. Eight are set by this project where no general standard exists; these are marked "Project".

Group	Criterion	What it measures	Usual pass value	Source
Cell quality	Inverted cells	Cells with negative volume	0	checkMesh fails any mesh with one
Cell quality	Minimum scaled Jacobian	Worst cell shape: 1 = perfect cube, 0 or less = inverted	> 0; this pipeline enforces ⩾ 0.05	Project (guards in step 3b and step 4)
Cell quality	Max non-orthogonality	Angle between a face normal and the line joining the two cell centres	⩽ 70° (checkMesh default); 40–70° needs 1 non-orthogonal corrector	OpenFOAM primitiveMeshCheck.C, CFD Pilot
Cell quality	Max skewness (OpenFOAM definition)	Distance of a face centre from the line joining the two cell centres	⩽ 4 (checkMesh default); 2–4 acceptable for most solvers	OpenFOAM primitiveMeshCheck.C, CFD Pilot
Cell quality	Max aspect ratio	Longest to shortest cell dimension	⩽ 1000 (checkMesh default); 100–10,000 is normal in wall-layer cells	OpenFOAM primitiveMeshCheck.C, CFD Pilot
Cell quality	checkMesh verdict	All standard topology and geometry checks	"Mesh OK" (0 failed checks)	OpenFOAM checkMesh
Cell quality	Orthogonal quality (Fluent)	Fluent's combined orthogonality measure, 1 = best	Minimum > 0.1 (best practice); 0.01 absolute minimum	Ansys forum

Group	Criterion	What it measures	Usual pass value	Source
Cell quality	Skewness (Fluent, equiangle)	Deviation from an ideal cell shape, 0 = best	Max < 0.95, average < 0.33	Ansys Fluent 12 User's Guide, 6.2.2
Wall resolution	First-cell y+	Height of the first cell in wall units	< 1 for wall-resolved k-ω SST (target 0.5); < 2 acceptable for engineering accuracy	CFD Handbook, k-ω SST
Wall resolution	Growth ratio	Height ratio of neighbouring cells	Change < 20 % between neighbours (ratio ⩽ 1.2)	Wolf Dynamics, mesh quality
Wall resolution	Layer coverage	Share of the wall area that carries the full layer stack	⩾ 99 %	Project (limit built into step 3b)
Geometry	Surface deviation	Distance of mesh wall nodes from the STL surface	Mean ⩽ 0.25 × atom size; max ⩽ 1 × atom size	Project
Geometry	Volume deviation	Mesh volume against STL volume	⩽ 1 % fluid; ⩽ 2 % solid	Project
Geometry	Interface area match	Fluid-side against solid-side interface area (CHT only)	⩽ 1 %	Project (step 4, MATCH_AREA_TOL)
Solution	Energy balance	Heat carried out by the water against heater power	Within 1 %	Project

Group	Criterion	What it measures	Usual pass value	Source
Solution	Interface heat error	Heat on the fluid side of the interface against the solid side	Within 1 %	Project
Solution	Mass balance	Outlet against inlet mass flow	< 0.1 %	Project

The OpenFOAM and Fluent skewness values use different definitions and cannot be compared directly. OpenFOAM is the solver here, so its limits decide pass or fail; the Fluent limits apply only if the Fluent export (.msh) is used.

Cell quality per design

All 20 regions (10 meshes, fluid and solid) pass checkMesh with "Mesh OK". No mesh has an inverted cell, and every cell has a scaled Jacobian of at least 0.050 (pass value > 0). The solids listed are the final ones: the pin-fin `_M` solids and the final-code TPMS solids.

Design	Mesh	Cells fluid / solid (M)	Min scaled Jacobian fluid / solid	Fluid cells with J < 0.1	Inverted cells
Chevron	Pin-fin, 0.3 mm, layers, matched interface	5.11 / 1.14	0.075 / 0.238	83	0
Cylinder	Pin-fin, 0.3 mm, layers, matched interface	5.11 / 1.49	0.060 / 0.203	22	0
Parallel	Pin-fin, 0.3 mm, layers, matched interface	4.93 / 1.68	0.060 / 0.443	34	0
Square	Pin-fin, 0.3 mm, layers, matched interface	4.77 / 1.37	0.070 / 0.308	91	0
Gyroid	TPMS, 0.2 mm, layers, matched interface	12.40 / 4.04	0.050 / 0.200	15,629	0
Lidinoid	TPMS, 0.2 mm, layers, matched interface	14.17 / 4.38	0.050 / 0.203	40,762	0
Neovius	TPMS, 0.2 mm, layers, matched interface	12.40 / 4.09	0.050 / 0.205	17,719	0
Gyroid	TPMS, 0.2 mm, no layers	6.25 / 4.03	0.263 / 0.289	0	0
Lidinoid	TPMS, 0.2 mm, no layers	5.92 / 4.36	0.267 / 0.280	0	0
Neovius	TPMS, 0.2 mm, no layers	6.21 / 4.08	0.268 / 0.286	0	0

Results of checkMesh (OpenFOAM v2312, standard checks). The pin-fin solids are the final `_M` meshes. The TPMS solids were checked on 5 October, before their outer walls were moved onto the CAD; the final-code TPMS solids passed the mesher's own export check (min scaled Jacobian 0.200–0.205, no cell below 0.1), and the gyroid ran with its final solid without problems. The final-code gyroid solid of the wall-layer test passes checkMesh with a non-orthogonality of 25.9°, skewness 1.92 and aspect ratio 3.40.

Design	Max non-orthogonality fluid / solid (°), limit 70	Max skewness fluid / solid, limit 4	Max aspect ratio fluid / solid, limit 1000	Verdict
Chevron	53.8 / 29.1	3.51 / 1.37	47.3 / 2.60	Mesh OK
Cylinder	57.2 / 28.9	3.44 / 1.56	44.7 / 2.63	Mesh OK
Parallel	50.7 / 29.3	3.53 / 1.22	39.3 / 2.49	Mesh OK
Square	49.9 / 29.1	3.56 / 1.35	47.9 / 2.55	Mesh OK
Gyroid, layers, matched	67.0 / 18.9	3.56 / 0.79	73.6 / 2.25	Mesh OK
Lidinoid, layers, matched	69.5 / 20.4	3.56 / 0.78	102.9 / 2.29	Mesh OK
Neovius, layers, matched	69.4 / 22.3	3.56 / 0.81	78.5 / 2.22	Mesh OK
Gyroid, no layers	22.4 / 21.1	0.80 / 0.76	2.20 / 2.29	Mesh OK
Lidinoid, no layers	26.2 / 22.7	0.79 / 0.76	2.31 / 2.31	Mesh OK
Neovius, no layers	21.6 / 20.3	0.77 / 0.77	2.29 / 2.40	Mesh OK

The solids and the meshes without layers sit far inside every limit: non-orthogonality at most 30°, skewness at most 1.56, aspect ratio below 2.7. The final pin-fin solids score a little higher than the earlier, separately meshed ones (non-orthogonality about 29°

instead of 19°, skewness 1.2–1.6 instead of below 1), because their nodes follow the fluid's interface and the CAD outer walls rather than their own grid. The layered fluids are the demanding case. Their worst values all come from wall-layer cells:

- Skewness reaches 3.44–3.56, within the limit of 4. It falls in the 2–4 band described as acceptable for most solvers.
- Aspect ratio reaches 39–103. This is normal for thin wall layers and a tenth of the limit.
- Non-orthogonality reaches 50–57° in the pin-fins and 67.0–69.5° in the TPMS meshes. Lidinoid and Neovius are within 1° of the 70° limit. The averages are only 4.9–8.1° (pin-fins) and 10.4–13.9° (TPMS), so few faces are affected. In the 40–70° band the guidance asks for at least one non-orthogonal corrector. The pin-fin runs use 1 and the TPMS runs use 2.

All low-Jacobian cells ($J < 0.1$) are thin wall-layer cells on curved walls. Meshes without layers stay at 0.26 or above. The solids stay at 0.20 or above, because the outer-wall correction is not allowed to push a cell below 0.2. In the TPMS layered meshes the cells below 0.1 are 0.25–0.49 % of the layer cells; in the pin-fins, fewer than 100 cells per mesh.

The layered TPMS fluids have about twice the cells of the no-layer fluids because the 5 layers add 6.1–8.3 M cells. Lidinoid has the most because it has the largest wetted area.

Wall resolution

Every layered mesh meets the y+ target: mean y+ is 0.40–0.60 (pass value < 1). Layer coverage is 99.8–100 % (pass value ⩾ 99 %). The meshes without layers have mean y+ about 10. They fail the wall-resolved criterion, as expected; they serve as the reference that shows what the layers add.

Design	Layers	First layer (µm)	Growth ratio	Total layer height (µm)	Layer coverage	y+ mean	y+ max
Chevron	5	13.3	1.30	120	100 %	0.49	4.88
Cylinder	5	13.3	1.30	120	100 %	0.60	5.25
Parallel	5	13.3	1.30	120	100 %	0.40	4.13
Square	5	13.3	1.30	120	100 %	0.58	4.73
Gyroid, layers	5	8.8	1.30	80	100 %	0.58	4.47
Lidinoid, layers	5	8.8	1.30	80	99.835 %	0.59	30.2
Neovius, layers	5	8.8	1.30	80	100 %	0.57	4.55
Gyroid, no layers	0	200 (atom)	–	–	–	10.1	39.3
Lidinoid, no layers	0	200 (atom)	–	–	–	9.9	43.8

Design	Layers	First layer (µm)	Growth ratio	Total layer height (µm)	Layer coverage	y+ mean	y+ max
Neovius, no layers	0	200 (atom)	–	–	–	10.3	40.4

y+ is measured on the fluid side of the interface at the end of the run. The layers are built inside the wall atom, and their total height is capped at 0.4 × atom size. This cap scaled the planned 20 µm first layer to 13.3 µm (pin-fins) and 8.8 µm (TPMS).

Three points do not meet the published guidance:

- **Maximum y+ of 4–5.** It is above 1 on a small share of faces in all layered meshes, while the mean stays near 0.5. The guidance asks for < 1, with < 2 acceptable, so the peaks are a minor departure.
- **Lidinoid maximum y+ of 30.** This is most likely at the 0.165 % of wall area where layers were removed to keep cells valid. There, the first cell is a full atom, as in the no-layer meshes.
- **Growth ratio of 1.30.** This is above the 1.2 guideline. The larger step is from the last layer to the first core cell. That cell is about 0.6 atom thick (180 µm pin-fin, 120 µm TPMS), so the jump is about 4.7× (37.9 → 180 µm and 25.3 → 120 µm).

The energy balance and y+ results show that the solutions are not harmed. A reviewer applying the usual rules will still flag the transition, so a wall-layer test with a smoother transition was run; see Limits and open checks.

Geometric fidelity

All meshes reproduce the CAD geometry within the project limits. The mean fluid wall deviation is 4–20 % of the atom size (limit 25 %), and the solid's outer walls are within about 2 µm of the CAD. Fluid volume is within 0.2 % (limit 1 %) and solid volume within 0.32 % (limit 2 %).

Design	Atom (mm)	Fluid wall deviation mean / max (µm)	Solid outer walls, mean (µm)	Volume deviation fluid / solid (%)
Chevron	0.3	18.6 / 151.1	1.7	−0.06 / +0.16
Cylinder	0.3	20.2 / 178.7	1.7	−0.20 / +0.32
Parallel	0.3	19.8 / 175.7	2.2	−0.09 / +0.14
Square	0.3	12.8 / 154.2	1.8	+0.02 / −0.00

Design	Atom (mm)	Fluid wall deviation mean / max (µm)	Solid outer walls, mean (µm)	Volume deviation fluid / solid (%)
Gyroid, layers, matched	0.2	35.7 / 146.3	1.6	+0.04 / −0.04
Lidinoid, layers, matched	0.2	39.2 / 160.2	1.7	+0.12 / −0.12
Neovius, layers, matched	0.2	33.9 / 148.0	1.7	+0.04 / −0.03
Gyroid, no layers	0.2	29.4 / 121.3	12.5 (all solid walls)	−0.03 / +0.03
Lidinoid, no layers	0.2	30.0 / 116.8	16.5 (all solid walls)	−0.05 / +0.06
Neovius, no layers	0.2	28.4 / 123.7	11.8 (all solid walls)	−0.02 / +0.05

Fluid deviation is the distance from each fluid wall node to the STL surface, over all fluid walls. The solid's outer walls are its walls that do not touch the fluid. Its interface nodes take the fluid's positions, so on the interface the solid has the fluid's deviation. For the no-layer TPMS meshes (separately meshed solids) the value is over all solid walls. The limits scale with the atom size: a mean of 75 µm and a max of 300 µm at 0.3 mm, and 50 µm and 200 µm at 0.2 mm.

The tightest margin is the maximum deviation on the TPMS solids: 183 µm, 92 % of the 200 µm limit. It sits at a few outer-wall nodes where the cell-quality guard reduced the correction. 99 % of the outer-wall nodes are within 35 µm, and the mean is 1.6–1.7 µm. Before the correction, the outer walls of the shared-grid solids were on average 24 µm (TPMS) to 41 µm (pin-fins) from the CAD.

The pin-fin solids are within 0.32 % of the CAD volume. When they were meshed separately, they were 1.0–1.6 % too large, because the 0.3 mm atoms overfilled the thin fins and corners.

Interface area match (CHT). On the pin-fins, 100 % of the interface faces are identical on both sides, and the areas agree to 0.00 %. On the matched TPMS meshes, 99.0–99.5 % of the faces are identical, and the areas agree to −0.06, −0.05 and +0.02 % (gyroid, Lidinoid, Neovius). Before the shared grid, the TPMS fluid side was larger by 3.58, 5.14 and 3.12 % (measured in OpenFOAM), which failed the 1 % criterion.

Solution-level checks

With the final meshes, every pin-fin and layered TPMS run closes the energy balance within 0.13 %, the pin-fins within 0.01 %. The only run outside 1 % is the Lidinoid without layers (98.65 %), which still has a separately meshed solid. The first pin-fin runs, before the solid was meshed on the fluid grid, are listed for comparison.

Design	Heater power to water (%)	Interface heat error (%)	Fluid balance (%)	Pass (within 1 %)
Chevron, final	99.996	0.000	99.996	Yes
Cylinder, final	99.996	0.000	99.996	Yes
Parallel, final	99.993	0.000	99.993	Yes
Square, final	99.991	0.000	99.991	Yes
Gyroid, layers, final	100.027	+0.036	99.992	Yes
Lidinoid, layers, matched (5 Oct)	100.12	+0.13	99.996	Yes
Neovius, layers, matched (5 Oct)	100.03	+0.04	99.992	Yes
Gyroid, no layers	99.23	−0.77	99.99	Yes
Lidinoid, no layers	98.65	−1.34	99.99	No
Neovius, no layers	99.17	−0.82	99.99	Yes
Chevron, first run (separate solid)	99.80	−0.19	99.99	Yes
Cylinder, first run	97.78	−2.21	99.99	No
Parallel, first run	100.10	+0.11	99.99	Yes
Square, first run	99.35	−0.64	99.99	Yes

Heater power to water = $\dot{m} \cdot cp \cdot (T_{outlet} - T_{inlet}) / 2000$ W, with cp = 4180 J/(kg·K) and T inlet = 293 K. Interface heat error = (fluid-side heat − solid-side heat) / solid-side heat. Fluid balance = heat to water / fluid-side interface heat. Values are from the end of the frozen-flow stage.

- Fluid and solid conserve energy on their own. The fluid balance is 99.99 % and the solid balance 99.9998–100.0000 % in every run. Any larger error therefore sits at the interface.

- Interface errors follow interface conformity. Separately meshed interfaces (first pin-fin runs, no-layer TPMS) are off by 0.1–2.2 %, because the nearest-face mapping between non-matching faces is not conservative. Matched interfaces are within 0.13 %, and exactly zero where all faces are identical (pin-fins). The first layered TPMS runs, before the fix, gained 4.70, 7.39 and 3.72 %; this is the check that exposed the area mismatch.
- Convergence. The energy residual falls to about 1×10^{-10} in the frozen stage (final runs $0.96–2.2 \times 10^{-10}$). Mass flow is conserved to about 1×10^{-7} relative in the pin-fin runs, far below the 0.1 % limit. Mass balance for the TPMS runs has not been extracted.

Single-region fluid body and surface smoothing

An external fluid-only body (igniter burner, 9.4 × 41.9 × 9.4 mm, one walls patch) and the new optional surface smoothing (step 3c) pass the standard checkMesh in OpenFOAM v2312. These meshes were not simulated.

Mesh	Cells	Max non-orthogonality (average)	Max skewness	Max aspect ratio	Wall distance to CAD, mean / max (μm)	checkMesh
burner_02, 0.2 mm	409,144 (400 prism / wedge)	61.6° (10.6°)	3.89	84.1	30.1 / 147	Mesh OK
burner_01, 0.1 mm	2,513,452 hex	66.5° (8.4°)	3.52	85.1	14.8 / 63	Mesh OK
burner_01s, 0.1 mm, step 3c with a cell-only guard	2,513,452 hex	68.4° (9.4°)	4.14 (884 faces)	73.2	1.6 / 42	Failed 1 check
burner_01s2, 0.1 mm, step 3c with the face guard	2,513,452 hex	68.4° (9.4°)	3.80	73.2	1.6 / 42	Mesh OK

Step 3c now limits checkMesh's skewness to 3.8 and non-orthogonality to 69° for every face it changes, computed with checkMesh's formulas; burner_01s2 stopped at 3.79999. On the layered test body, step 3c changed the `-allGeometry` counts as follows: face tets 11,337 → 14,331 faces, concave cells 29 → 96, small-determinant cells 4,909 → 806. Without wall layers, the cells behind the wall absorb the moves: non-orthogonality 23.7° → 53.4°, skewness 0.90 → 2.60.

Atom-test meshes

The square pin-fin at 0.45 and 0.2 mm passes checkMesh in both regions; the gyroid at 0.15 mm fails it in the solid on skewness. All three keep identical interface faces (99.34–100 %). They were meshed with the final code and default settings for the atom test (26–29 October).

Mesh	Fluid cells	Solid cells	Max skewness, fluid / solid	Max aspect ratio, fluid / solid	Min scaled Jacobian, fluid / solid	Fluid / solid volume vs CAD	checkMesh
Square 0.45 mm (square_mi045)	1,726,158	465,988	3.40 / 1.85	48.7 / 2.34	0.066 / 0.319	−1.34 % / +2.27 %	Mesh OK, both
Square 0.3 mm (square_M, reference)	4,771,032	1,373,560	3.56 / 1.35	47.9 / 2.55	0.070 / 0.308	+0.02 % / −0.00 %	Mesh OK, both
Square 0.2 mm (square_mi02)	13,547,700	4,970,660	3.42 / 3.47	39.4 / 3.46	0.101 / 0.385	+0.05 % / −0.08 %	Mesh OK, both
Gyroid 0.15 mm (gyroid_mi015)	25,985,046	9,380,944	3.53 / 4.61	87.2 / 3.98	0.050 / 0.201	−0.03 % / +0.06 %	Fluid OK; solid failed (522 faces above skewness 4)

- The solid's maximum skewness rises as the atom shrinks (1.35 at 0.3 mm, 3.47 at 0.2 mm on the square, 4.61 at 0.15 mm on the gyroid), although its minimum scaled Jacobian stays at 0.20 or above. The cause, most likely the outer-wall placement in Step 4 at small atoms, is not yet known. The gyroid run on this solid behaved normally (interface heat error +0.019 %, 99.9 % of the heater power to the water).
- At 0.45 mm the square misses the volume limits (fluid −1.34 % against 1 %, solid +2.27 % against 2 %), and its mean fluid wall distance is 49 µm (max 295 µm). It is a coarse test grid, not a mesh for use.
- Gyroid 0.15 mm fluid: 171 layer cells failed the guard (0.002 % of the layer cells), so layers were removed at 102 wall nodes.
- Non-orthogonality was not extracted from these checkMesh logs.

The Qu & Mudawar meshes (0.0483 / 0.0342 / 0.0239 mm, 1.10–9.96 M cells) were made in the web app on a PC; their checkMesh logs were not collected on the cluster.

Limits and open checks

Six items remain. None of them blocks the current results, but each should be closed or explained before an industry review. Closed since the last version: the interface heat error of the separately meshed pin-fin solids (now 0.000 %), the cross-code check (SimScale comparison done; see the validation report), and the layer-to-core jump (wall-layer test, first item below).

- **Growth ratio and layer-to-core jump.** The ratio is 1.30 against a 1.2 guideline, and the jump to the core cell is about 4.7×. The wall-layer test (`layer_test.sh`, 13 October) used 6 layers with a total height of up to 0.6 × atom size, which lowers the jump to about 2.3, on the square pin-fin and the gyroid. The peak heater temperature changed by −0.61 K and +0.13 K, within the oscillation bands of 0.67 K and 0.23 K. The pressure drop changed by +0.0002 kPa on the square (band 0.34 kPa) and +1.16 kPa on the gyroid (band 1.06 kPa; +0.90 kPa as the mean of iterations 2500–4000). The heater average temperature fell by 0.25–0.39 K. All four test regions pass checkMesh; on the gyroid the guard thinned the layers at 14.6 % of the wall nodes, and the fluid cells below J = 0.1 rose from 15,629 to 32,955. The default stays at 5 layers and a cap of 0.4, and the layer setting is reported as an uncertainty: 0.6 K in peak and 0.4 K in average heater temperature, 0.1 kPa in pin-fin and 1.2 kPa in TPMS pressure drop. Status: closed, with this uncertainty.
- **Solid skewness at small atoms.** The 0.15 mm gyroid solid failed checkMesh on skewness (4.61, 522 faces), and the square solid reached 3.47 at 0.2 mm (section Atom-test meshes). The mesh-independence test itself is done; see the validation report.
- **Non-orthogonality margin.** Lidinoid (69.5°) and Neovius (69.4°) pass but sit within 1° of the 70° limit, so a finer atom or a different layer setting could tip a new design over it. Option: let the step 3b guard, which already removes layer cells above 70°, use a stricter limit such as 65°, and check the effect on layer coverage.
- **No-layer TPMS meshes.** These reference runs still have separately meshed solids and lose 0.8–1.3 % at the interface. Meshing their solids on the fluid grid (now the default in `mesh_design.sh`) would remove this. They are kept as they are because they serve only as the no-layer reference.
- **Fluent quality metrics.** Orthogonal quality and equiangle skewness have not been measured. The Fluent export is now off by default (`MESH_WRITE_FLUENT=0`); the metrics are needed only if it is offered to users.
- **Face tets.** checkMesh -allGeometry lists some layer faces with low-quality face-tet decomposition (for example 5,216 of 85,440 gap-edge faces in Lidinoid). The standard checkMesh does not count this as a failure, and the pipeline does not enforce it (STRICT_FACE_TETS off).

- **Surface smoothing (step 3c).** Checked by checkMesh and the CAD distance only; its effect on wall shear, heat transfer and pressure drop is not measured. It leaves the CHT interface locked, so the campaign's CHT meshes are unaffected. Planned: a laminar pipe or bend with and without step 3c against the Hagen–Poiseuille pressure drop.

The Schwarz TPMS design is not part of this report. Its fluid STL contains 32 sealed pockets (14.6 % of the fluid volume), so the geometry, not the mesh, is invalid.

Sources

- OpenFOAM primitiveMeshCheck.C source (default thresholds)
- CFD Pilot: OpenFOAM checkMesh – how to read and fix mesh quality errors
- Wolf Dynamics: meshing preliminaries and quality assessment
- Ansys Fluent 12.0 User's Guide, 6.2.2 Mesh Quality
- Ansys Innovation Space forum: good orthogonal quality values
- CFD Handbook: k-ω SST model

Project data: mesh logs of the pin-fin runs (3 Oct 2026), the TPMS `_L`, `_NL` and `_LM` runs (4–5 Oct), the final solid exports (pin-fin `_M` and the TPMS check exports, 5 Oct), checkMesh of all regions, and the solver results of the final runs (8 Oct). All values are in validation_campaign.xlsx.